# Homework 09

Steven Jia

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## Question 1: Choosing an SVM kernel

I used the 462 observations in this dataset to predict `chd`, using the same nine predictors as before. I coded `famhist` as Absent = 1 and Present = 2. I compared 18 settings. For each kernel, I tried $C = 0.1, 1, 10$. For polynomial kernels, I used degrees 2, 3, and 4, with $\gamma = 1/9$ and coef0 = 1 fixed. For radial, I also tried $\gamma = 0.01, 0.1$.

Then I set the random seed to 43209, shuffled observations within each class, and divided them into five folds. All models used the same folds. In each fold, I calculated the means and SDs using only the training part, then used them to standardize the validation part. The table shows the average of the five fold error rates.

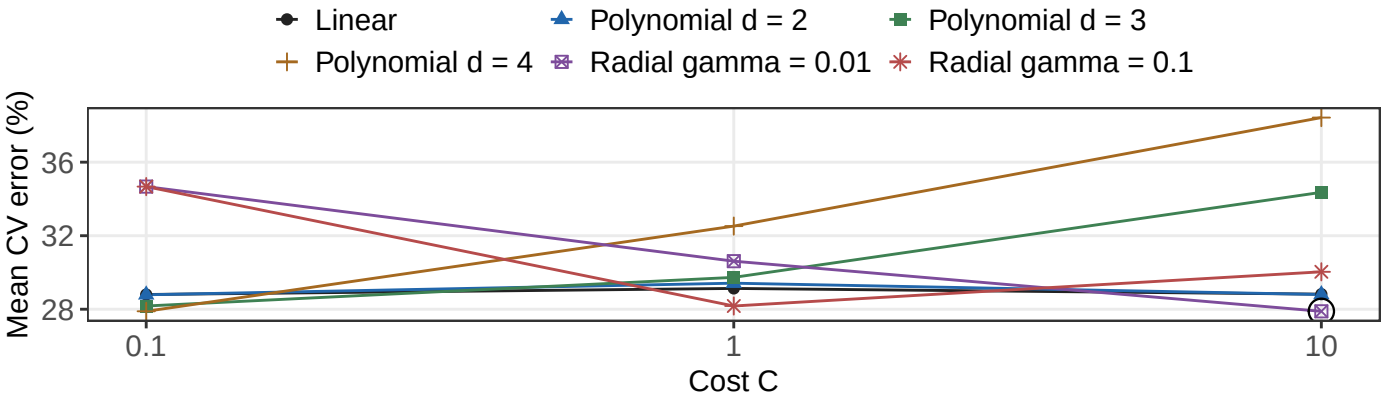
| Kernel | C | Gamma | Mean CV error (%) |
|---|---|---|---|
| Linear | 0.1 | – | 27.73 |
| Linear | 1.0 | – | 29.24 |
| Linear | 10.0 | – | 29.24 |
| Polynomial, d = 2 | 0.1 | 1/9 | 28.36 |
| Polynomial, d = 2 | 1.0 | 1/9 | 29.23 |
| Polynomial, d = 2 | 10.0 | 1/9 | 30.32 |
| Polynomial, d = 3 | 0.1 | 1/9 | 27.72 |
| Polynomial, d = 3 | 1.0 | 1/9 | 31.40 |
| Polynomial, d = 3 | 10.0 | 1/9 | 33.35 |
| Polynomial, d = 4 | 0.1 | 1/9 | 29.24 |
| Polynomial, d = 4 | 1.0 | 1/9 | 33.34 |
| Polynomial, d = 4 | 10.0 | 1/9 | 34.87 |
| Radial | 0.1 | 0.01 | 34.63 |
| Radial | 1.0 | 0.01 | 27.07 |
| Radial | 10.0 | 0.01 | 27.28 |
| Radial | 0.1 | 0.1 | 34.42 |
| Radial | 1.0 | 0.1 | 29.23 |
| Radial | 10.0 | 0.1 | 31.84 |

Here I chose the radial kernel with $C = 1$ and $\gamma = 0.01$, mainly because its average error rate was the lowest, **27.07%**. The best linear model had an error rate of 27.73%, a difference of about 0.65 percentage points. So I think radial was a reasonable choice in this comparison.

## Question 2: SVM analysis report

This dataset has 462 observations: 302 without CHD and 160 with CHD. The nine predictors have no missing values, and there are no duplicate observations or constant columns. I coded `famhist` as Absent = 1 and Present = 2. I think SVM can be used for this binary classification problem.

Then I split the data by class into 323 training and 139 test observations, using seed 43209. I compared 18 settings using five-fold cross-validation on the training set, with seed 43210 for the folds. Standardization used only the training part of each fold. Linear provides a simple boundary first, while polynomial and radial let me see whether nonlinear models work better.




| Best setting by kernel | C | Gamma | Mean CV error (%) |
|---|---|---|---|
| Linear | 0.1 | – | 28.80 |
| Polynomial, degree 2 | 0.1 | 1/9 | 28.79 |
| Polynomial, degree 3 | 0.1 | 1/9 | 28.18 |
| Polynomial, degree 4 | 0.1 | 1/9 | 27.89 |
| Radial | 10.0 | 0.01 | 27.89 |

In the end, I chose radial with $C = 10$ and $\gamma = 0.01$, mainly because its average five-fold error rate was the lowest. Its error rate and the degree-4 polynomial error rate both round to **27.89%**. The actual difference was very small.

The model misclassified **38 of the 139 test observations**, giving an error rate of **27.34%**, with a 95% Wilson interval of **20.61–35.28%**. Among people without CHD, **87.91%** were classified correctly; among people with CHD, only **43.75%** were classified correctly. If everyone were predicted as class 0, the largest class in the training set, the test error rate would be **34.53%**.

| Observed | Predicted 0 | Predicted 1 |
|---|---|---|
| 0 | 80 | 11 |
| 1 | 27 | 21 |

I found that **27 of the 48 CHD cases were missed**, which shows that the overall error rate alone is not enough. Both classes used the same cost, without an extra penalty for missing CHD cases. These results also depend on one data split, so another sample may not give the same results.

## Appendix: Complete SVM skill

```
---
name: svm-report
description: Fit and compare SVM classifiers and write a short report only when this skill is
 ↪  explicitly invoked.
---

# SVM report

Use this skill only when explicitly invoked. Receive the data, response,
predictors and coding in the invocation, not in this file.

- Use R and save runnable code, seeds, row IDs, settings and package versions.
- Check the response, predictors, missing values, duplicates, class counts
  and zero-variance columns. Explain whether SVM classification is suitable.
- Before fitting, keep about 30% of each class as an untouched test set.
  Use seed 43209 by default. Save the split before running any comparisons.
- Start with linear, polynomial degrees 2-4, and radial kernels. Explain
  which can be useful. Set the whole grid before seeing model results.
- By default, try C = 0.1, 1, 10 for every kernel. For polynomial kernels,
  fix gamma = 1/p and coef0 = 1, where p is the predictor count. For radial
  kernels, try gamma = 0.01, 0.1. This gives 18 settings. Use equal costs
  for both classes. Explain any changes before fitting; do not tune on test data.
- Assign five stratified CV folds on the training set, using seed 43210.
  Use the same folds for all settings. Reduce folds if a class is too small.
- In each fold, estimate centering, SD scaling and any other preprocessing
  from fitting rows only. Apply them to validation rows; avoid scaling twice.
- Select the lowest mean fold misclassification rate. For exact ties, prefer
  linear, then lower polynomial degree, then radial; then smaller C and gamma.
  Save every candidate's fold errors. Record failed fits; do not hide them.
- Save the selected setting before opening the test results. Refit it on all
  training rows and evaluate it once. Do not change it after this evaluation.
- Report test errors as count/total and percent, a 95% Wilson interval,
  and a confusion matrix with observed rows and predicted columns. Give each
  class's recall and the training-majority baseline. Explain missed cases
  when relevant. Do not treat the selected CV minimum as unbiased test error.
- Write at most one printed page at 11-point font. Include one compact CV
  figure and a table with the best setting from each kernel. Say clearly why
  the final model was selected, give its key results and a short limitation.
- Keep the full grid, fold assignments, code, invocation and complete skill
  outside that page. Use short, plain sentences; check numbers and page size.
```

### Skill changes, invocation, and checks

I read the review of my Homework 7 skill and the review I wrote in Homework 8. So I shortened the skill to 35 lines and kept only the main results in the report. I also stated the default settings and the model selection rule clearly, and saved the code and data splits actually used. This makes the analysis easier to repeat.

**Explicit invocation (data instructions; full run prompt in the archive):**

> Use `$svm-report` at `svm-report/SKILL.md` to analyze `data/saheart.csv`. Use `chd` as the response and `sbp`, `tobacco`, `ldl`, `adiposity`, `famhist`, `typea`, `obesity`, `alcohol`, and `age` as predictors. Code `famhist` as Absent = 1 and Present = 2. Save the code and results.

I checked whether the models used the same training folds and whether standardization used only the corresponding training data. I also checked the error rates against the saved predictions. The model choice was saved before the test results were examined.

**Reproducible companion.** The accompanying R Markdown archive contains `q1-analysis.R`, `q2-run/q2-analysis.R`, the data, the skill and invocation, all candidate results, fold assignments, and predictions. Open `homework-09.Rmd` in RStudio to knit the report from the saved results. Run `render.R` to rerun both fixed analyses and rebuild the report. The second question's analysis page is also supplied as a separate PDF.